TDI HNA Coil Troubleshooting

1 TDI HNA Troubleshooting tips

The 3.0T TDI HNA coil is a 21 Element Coil. This coil connects exclusively to the P2 Port.

The coil comprises 3 major components, the Posterior, Anterior Face Bill and Adapter Block.

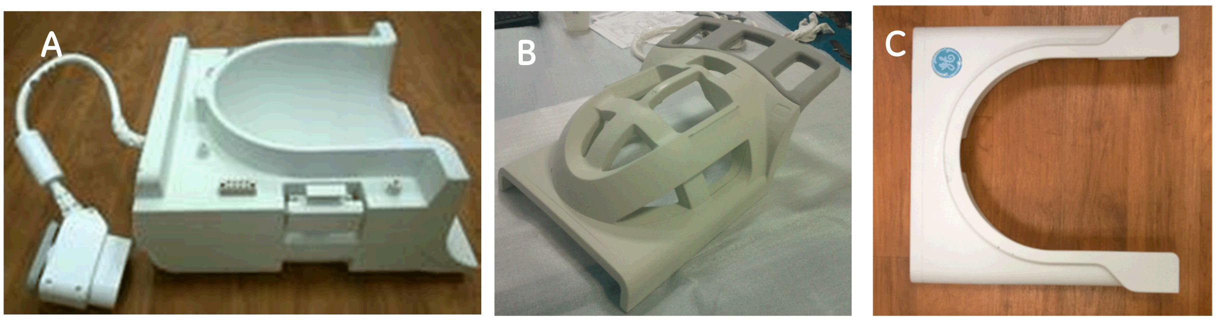
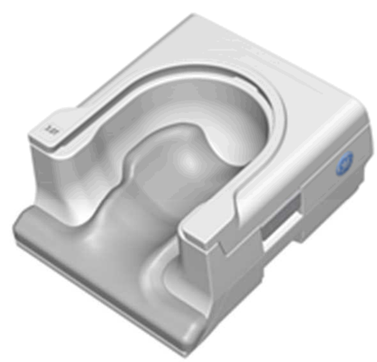


Fig 1: Coil Components: (A) Coil Posterior. (B) Coil Anterior Face Bill (C) Coil Adapter Block

These 3 components form the following 2 configurations of the TDI HNA Coil:

(i) Posterior + Anterior Face Bill: TDI HNU Face: 21 Element Coil with flexible anterior bill
(ii) Posterior + Adapter Block: 10 Element Coil using the Adapter Block.

Posterior with Anterior



Posterior with Adapter

This troubleshooting guide is to provide Service Engineers diagnostic tips for the most common failure mode possible in the 3.0T TDI HNA Coil.

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a. Failure Mode #1: Coil not recognized.

The 3.0T TDI HNU mode has 3 coil IDs that are used for the 3 different components of the coil.

Component Connected	***Coil ID***	***Coil Name Displayed***
Posterior Only	GBGLdNdtiMCr1zsLHQGLyUwzJq0kTKwy	TDI HNU Pos
Posterior + Anterior Face Bill	GBGLjDwiCLFvVnyjJx1KYLq8c1MAuB1y	TDI HNU Face
Posterior + Adapter Block	GBGLxfhcQH1KWWyf9yqcu9ALoM5HDquI	TDI HNU AB

Table 1: TDI HNU Coil ID Codes

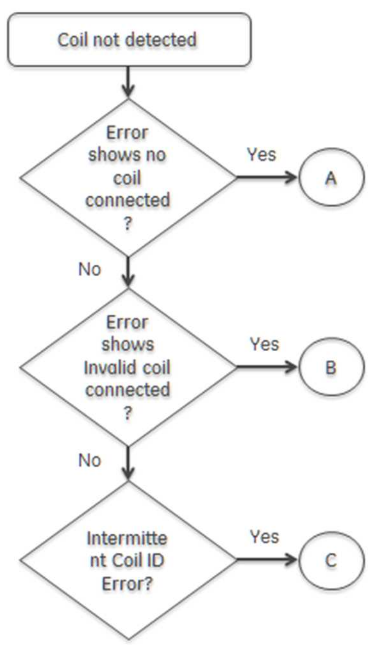
In normal use case, when only, the Posterior component of the coil is connected to P2 Port, '***TDI HNU POS***' coil name should be displayed on the 'In Room Monitor'. Similarly, when "Posterior+ Anterior Face Bill" or "Posterior + Adapter Block" is connected to P2 port, the 'In-Room Monitor' should display '***TDI HNU FACE***' and '***TDI HNU AB***' respectively.

Before diagnosing / replacing the coil, please run the following tests on system:

(i) 'Coil ID Functionality Functional diagnostics':
   a. Tool is located under' *Diagnostics > System Function > RF > Coil ID Functional Diagnostic*'

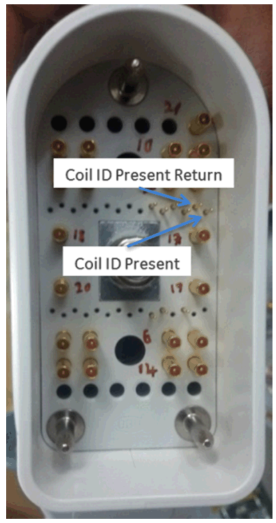
(ii) '*Coil Datapath Diagnostic*' Test: This will help eliminate the system as a source of Coil ID faults.
   a. Tool is located under 'Diagnostics >> System Function >> RF >> Coil Datapath Diagnostics.

If the system side is fine, please use the following flow chart should be used to diagnose 'Coil Not detected' fault for TDI HNU Coil:

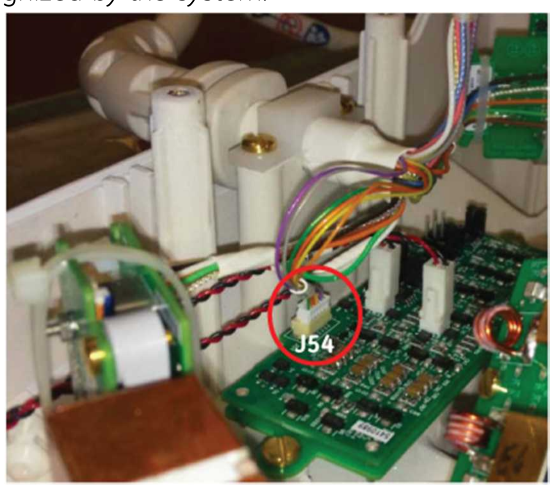


Scenario A:

(i) Check the Coil ID Present and Coil ID Present Pin on the P-Port Connector per image below. In case damage to the pins, please order the Cable FRU (Part#: 5566223)



(ii) If the pins are fine, please check J54 connector located on the Coil ID board, present on the inside the posterior of the coil. Reseat the connector and re-connect the coil to check if the coil is being recognized by the system.



If step (i) and (ii) do not fix the problem, the issue is related to the Coil ID board. Order the Coil Only FRU (Part#: 5882320)

Scenario B:

(i) If System shows 'Invalid Coil Connected' error. Please check the Coil ID Log file. The '*coilID.log*' file is located in '*/usr/g/service/log/*' folder of the system

(ii) Compare the values of the Coil ID listed in the log file, with the Coil ID details listed in the Table 1.

(iii) If the values in the log file do not match the codes listed above, please order the Coil FRU Part#: 5882320) as the issue is with the Coil ID board.

<u>*Scenario C:*</u>

(i) Intermittent Coil ID detection can be attributed to following 3 components:

a. P-Port Connector:
   i. Please check the pins highlighted in the image below. If any damage to any of the 6 pins, please order the Cable FRU assembly and re-test the coil.

b. Cable Assembly:
   i. Inspect the Coil cable assembly for any damage to cable, including Nicks and cuts to the cable.
   ii. If damage present, please order the cable FRU assembly

c. Coil ID Board Connector (J54)
   i. Please reseat the J54 connector of the coil ID Board and re-test the coil.
   ii. If there is damage to the connector or the intermittency persists, please order the Cable assembly FRU.

(ii) If the issue continues to persist, in spite of cable replacement, please order the Coil assembly FRU.

b. Failure Mode #2: MC Bias Fault:

The Multi Coil Bias failure usually occurs under the following condition:

- <u>Open circuit fault</u> - This fault is detected when multicoil bias current is applied, and the voltage generated is 80% of the rail voltage or greater. The rail voltage is +/- 10V. This fault can only be detected on a connected coil channel.
- <u>Short circuit fault</u> - This fault is detected when multicoil bias voltage is applied, and the current generated is greater than 70 mA. This fault can be detected on either a connected coil channel, or on an unconnected channel.

- Tolerance/drive fault - This fault is detected when multicoil bias voltage is applied, and the voltage read back is not within system tolerances, but only if a short circuit is not detected. This fault can be detected on either a connected coil channel, or on an unconnected channel.

To troubleshoot the multi-coil bias, run the Multi Coil Bias Diagnostic Test in the Service Browser. The tool is located at: '*Diagnostics > System Function > RF > Multicoil Bias Diagnostic*'